TruthLens 499B: Evidence-Grounded Multimodal Misinformation Verification for Bangla Social-Media Content

Interim Research Progress Report

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TABLE I
DATA EXTRACTION SUMMARY

Metric	Value	Metric	Value
Articles collected	2,603	Media URL records	7,580
Unique URLs	2,603	Image URLs	7,435
Extracted body	2,603 (100%)	Iframe URLs	144
Missing body	0	Video URLs	1

I. INTRODUCTION

This report presents the recent progress in expanding the dataset for Bangla fact-checking using articles from Rumor Scanner [1]. The primary objective is to collect high-quality fact-checking articles, extract relevant information such as article body, claim candidates, and verdict labels, and gather associated media (images and videos) for further analysis. The work builds a structured and verifiable dataset that can be used for developing automated fact-checking models in the Bangla language.

II. DATA COLLECTION AND PREPROCESSING

A total of 2,603 fact-check articles were collected from Rumor Scanner. Initial quality checks were performed to remove duplicates, validate URLs, and identify missing fields. The collected data includes article title, URL, publication date, and source verdict candidate. The dataset was then enriched with extracted claim candidates and verdict labels for each article.

III. HTML BODY EXTRACTION

To obtain clean and relevant article content, an automated HTML parsing pipeline was implemented [2], [3]. The system uses the CSS selector *.td-post-content* to extract the main article body from each webpage. The extracted text includes the full article content, character count, and extraction status. The extraction process was tested on 30 articles (*html_body_audit_30.csv*) before applying it to the entire dataset.

IV. MEDIA URL EXTRACTION

During the extraction process, all image and video URLs from the article body were collected. A total of 7,580 media URL records were identified, including 7,435 image URLs, 144 iframe URLs, and 1 video URL. The media files have not been downloaded yet and their relevance to the fact-check claims is under verification.

V. CURRENT PROGRESS

The dataset expansion process has been successfully completed for the initial stages. All 2,603 articles have been processed and their body text has been extracted without any missing content. The media manifest file (*media_manifest.csv*) has also been generated, containing all related image and video URLs. The extraction pipeline and intermediate results are stored in *factcheck_html_body_full.csv* for further analysis.

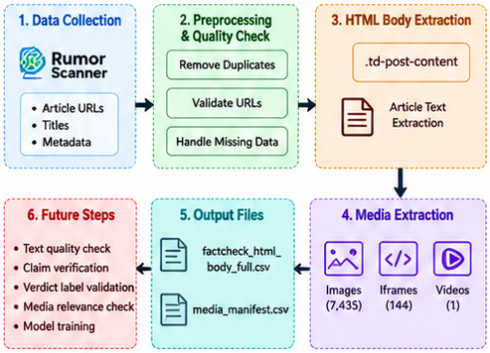


Fig. 1. Overall system architecture for dataset expansion and extraction.

VI. SYSTEM ARCHITECTURE

Fig. 1 shows the overall pipeline, from data collection to the planned future steps.

VII. NEXT STEPS

Although the initial extraction process is completed, several further steps are planned to improve the dataset quality and usability: (1) validate the extracted text for semantic correctness; (2) cross-check claim candidates and verdict labels with the original source; (3) verify the relevance and validity of extracted media URLs; (4) download and organize relevant images and videos; and (5) prepare the final verified dataset for model training and evaluation.

VIII. CONCLUSION

This report summarizes the collection and initial processing of 2,603 fact-check articles from Rumor Scanner. The HTML body extraction and media URL collection have been completed, providing a foundation for the next phase of the project. The current dataset is ready for further quality validation and model development.

REFERENCES

[1] Rumor Scanner, "Fact-Check Articles." [Online]. https://rumorscanner.com/
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[3] Python Requests, "HTTP for Humans." [Online]. https://docs.python-requests.org/